

blueprism®

P-017- Automated Password Strength Analysis and Reporting

Process Definition Document (PDD)

NOTE:

This version of the document and its current content is meant to serve as an example for business users (e.g., process SMEs) and it is intended to help with the creation of the process design documentation for RPA.

The current example`s content is fictive or adjusted to remove real confidential data and it should not be replicated in the automation of other business processes. All the process steps and screenshots in the PDD should be captured entirely from scratch and included here for the automation of the process scope.

The list of examples is not exhaustive. Additional entries may be added or removed, case by case, as required to provide relevant data for RPA.

Document History

Date	Version	Role	Name	Organization (Dept.)	Function	Comments
08 Aug 2025	1.0	Business Analyst	Doaa Ahmad	IT-RPA	IT	Initial draft of PDD

Document Approval Flow

Version	Flow	Role	Name	Organization (Dept.)	Approved Date:
1.0	Document prepared by	Business Analyst	Doaa Ahmad	IT-RPA	
1.0	Document Approved by:	Business Analyst	Nada AbuRub	IT-RPA	
1.0	Document Approved by:	Process Owner			

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I. Introduction

I.1 Purpose of the Document

The Process Definition Document outlines the business process chosen for automation using Blue Prism Automation (RPA) technology.

The document describes the sequence of steps performed as part of the business process, the conditions and rules of the process prior to automation and how they are envisioned to work after automating it, partly or entirely. This specifications document serves as a base for developers, providing them the details required for applying robotic automation to the selected business process.

I.2 Objectives

The Objective of the process is Preparing a Report.

The business objectives and benefits expected by the Business Process Owner after automation of the selected business process are:

- *Reduce processing time per item by 80 %*
- *Better Monitoring of the overall activity by using the logs provided by the robots.*

I.3 Key Contacts

The specifications document includes concise and complete requirements of the business process, and it is built based on the inputs provided by the process **Subject Matter Expert (SME)/ Process Owner**.

The **Process Owner** is expected **to review it and provide signoff for accuracy** and completion of the steps, context, impact, and complete set of process exceptions. The names have to be included in the table below.

Role	Name	Contact details (email, phone number)	Notes
Process SME	Sara Ababneh		
Process Reviewer	Sara Ababneh		
Process Owner/ Approver for production	Sara Ababneh		

1.4 Definition Table

The table contains abbreviations and their descriptions to clarify some of the abbreviations mentioned in the PDD, making them easier to understand and interpret.

Acronym	Description
FTEs	(Full-Time Employee): such as how many full-time employees are needed to complete certain tasks or phases of the product development process.
SLA	(Service Level Agreement): A formal contract or agreement that defines the level of service expected from a service provider. It outlines the specific performance metrics, responsibilities, and expectations for both the service provider and the client. Key elements typically include response times, availability, and performance standards. SLAs are critical in ensuring that services meet the required quality and reliability standards.
# of items processed /reference period	# of items processed / reference period: This metric represents the total number of items (such as tasks, products, transactions, etc.) that are completed, handled, or processed during a defined period (e.g., per hour, per day, per month).
Peak Period	Peak Period: A time interval during which the volume of activity, demand, or workload reaches its highest level. This period often requires additional resources, higher capacity, or enhanced performance to manage the increased demand effectively.
CAPTCHA	(Completely Automated Public Turing test to tell Computers and Humans Apart). It is a type of challenge-response test used in computing to determine whether the user is a human or a bot.
OTP	(One-Time Password). It is a security feature used to enhance the protection of user accounts and transactions by generating a unique, temporary password that is valid for only a single use or a limited time.
Identifier Field	An "Identifier Field" is a specific field in a database or system that contains unique values used to identify individual records or entries. This field typically ensures that each record is distinct and can be referenced accurately. Common examples include IDs, usernames, or any other unique identifier that differentiates one record from another in a dataset.
OCR	Optical Character Recognition (OCR) is a technology that enables the conversion of different types of documents, such as scanned paper documents, PDFs, or images captured by a digital camera, into editable and searchable data. OCR software analyses the patterns of text in an image and translates them into a digital format that can be manipulated, searched, or used in various applications.
UAT	User Acceptance Testing (UAT) in RPA is the final phase of testing where end users validate that the automated process meets business requirements and functions correctly in a real-world scenario. This ensures that the RPA solution delivers the expected outcomes before it goes live.
AI	Artificial Intelligence (AI) is the simulation of human intelligence by machines, enabling them to perform tasks such as learning, reasoning, problem-solving, and decision-making. It includes technologies like machine learning, natural language processing, and computer vision to automate and enhance processes.

I.4 Minimum Prerequisites for Automation

1. Filled in Process Design Document
2. Test Data to support development
3. User access and user accounts creations (licenses, permissions, restrictions to create accounts for robots)
4. Credentials (user ID and password) required to log on to machines and applications.
5. Dependencies with other projects on the same environment

II. As-Is Process Description

II.1 Process Overview

General information about the process selected for RPA prior to automation.

#	Item	Description
1	Process full name	Automated Password Strength Analysis and Reporting
2	Process Area	Cyborg Academy - Training Team
3	Department	Cyborg Academy - Training Team
4	Process short description (Operation, activity, outcome)	
5	Role(s) required for performing the process	Training Team
6	Process schedule and frequency	<i>daily</i>
7	# of items processes /reference period	<i>~10 requests</i>
8	Average handling time per item	<i>1 -2 min</i>
9	Peak period (s)	<i>N/A</i>
10	Transaction Volume During Peak period	<i>N/A</i>
11	Total # of FTEs supporting this activity	<i>1</i>
12	Expected increase of volume in the next reference period	<i>20%</i>
13	Level of exception rate	<i>20%</i>
14	Needs CAPTCHA	<i>N/A</i>
15	Needs OTP	<i>N/A</i>
16	Needs OCR	<i>N/A</i>
17	Need AI Tool	<i>N/A</i>
18	Input data	<i>Email</i>
19	Output data	<i>Reports via Email</i>

II.1 Applications Used in the Process

The table includes a comprehensive list all the applications that are used as part of the process automated, at various steps in the flow.

#	Application name & version	System Language	Environment/ Access method	User
1	Ms-Outlook	EN		
2	Kaggle	EN	https://www.kaggle.com/datasets/shivamb/10000-most-common-passwords	
3	bitwarden	EN	https://bitwarden.com/password-strength/	
4	Excel			

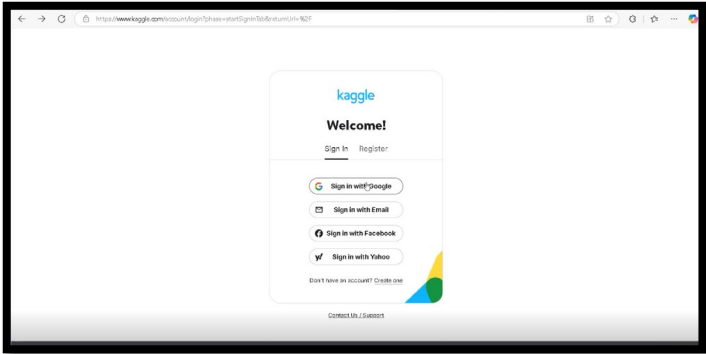
II.2 Detailed Process Map

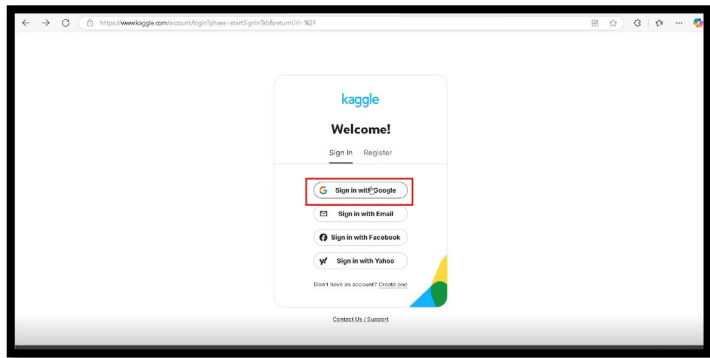
II.2 Detailed As is Process Steps

steps in the "As Is" are the same as the steps in the "To Be".

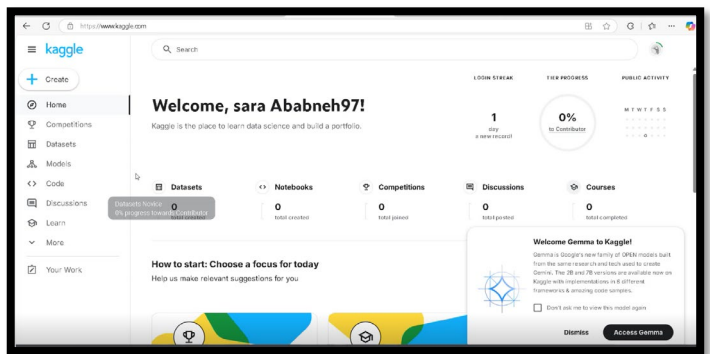
11.3 Detailed to be Process Steps

This chapter depicts the business process in detail to enable the Developer to build the automated process.

Step #	Step Description	Notes / Comments
Section 1: Receive input email		
1.	Once bot receive an email, it should start the process. 1.1. Email details: 1.1.1. Subject: 1.1.2. Body:	
Section 2: Access to Kaggle		
2.	Every day, Access to Kaggle portal. 2.1. Access to Kaggle portal, URL:" https://www.kaggle.com/datasets/shivamb/10000-most-common-passwords "  2.2. Click on Sugn in with Google button.	

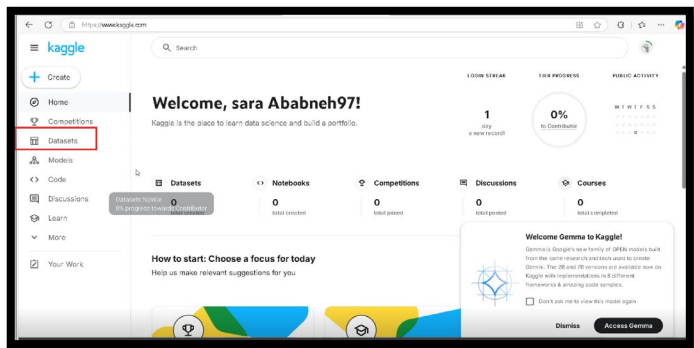


2.3. The screen will be as shown below:

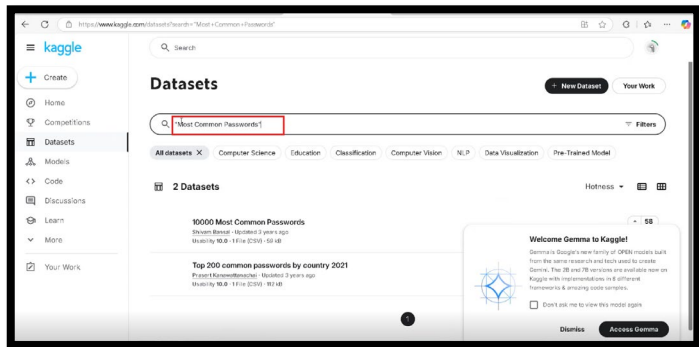


3. Access to must common password.

3.1. Click on **Datasets** icon.

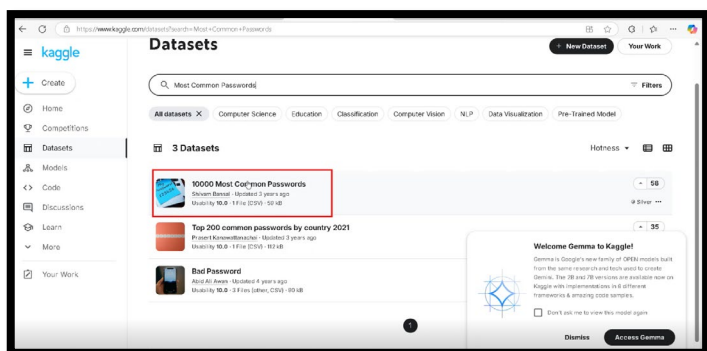


3.2. Enter **most common password** in **Search** field.

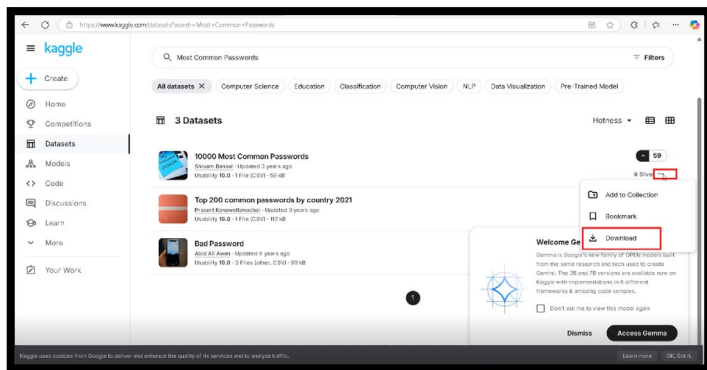


3.3. Click on **Enter**.

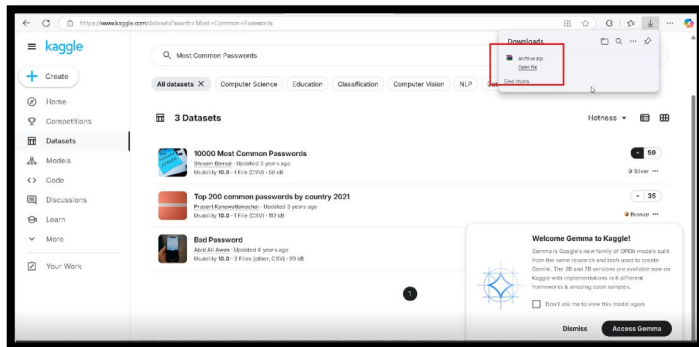
3.4. Click on Most common password.



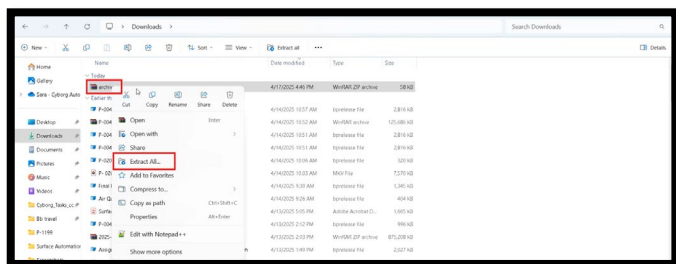
3.5. Download the attached.



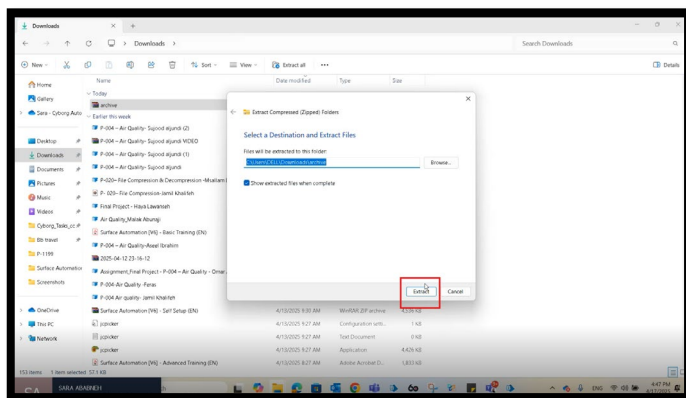
3.6. Open the downloaded file as below.



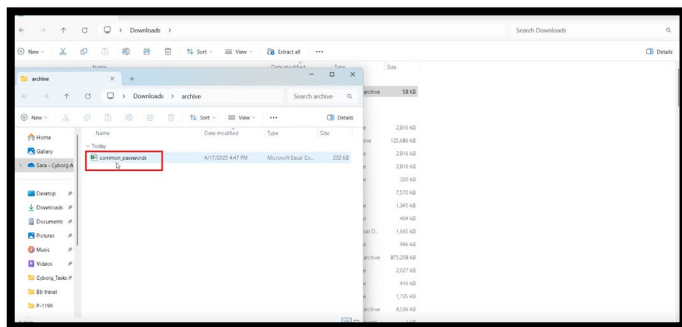
3.7. Right click, choose extract all



3.8. Click Extract button.



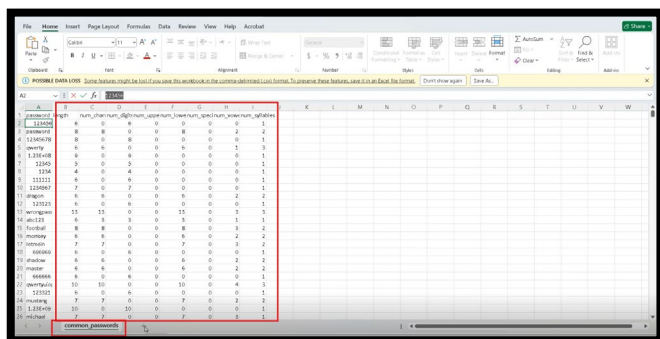
3.9. Open the common password excel sheet.



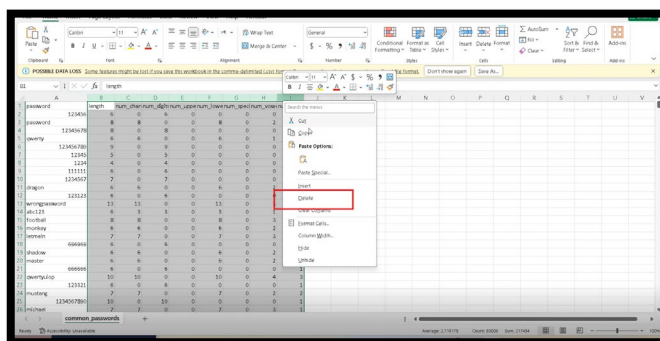
Section 3: Retrieve required data from common excel file

4. Manipulate the data to match with requirements.

4.1. select all columns exclude **password** columns.



4.2. Right click, then choose **Delete** option.



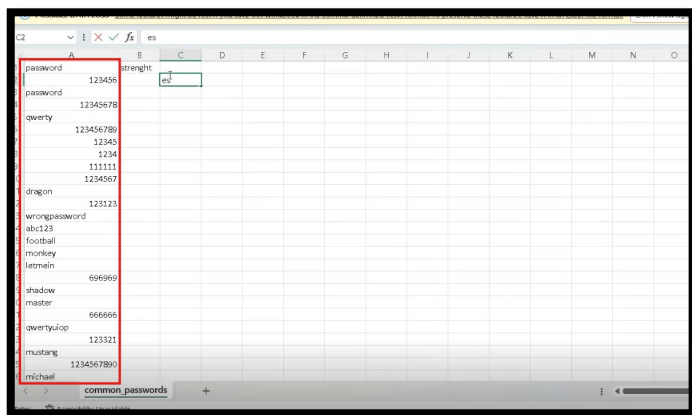
4.3. Add two new columns with names:

4.3.1. **Strength.**

4.3.2. **Estimated time.**

5. Retrieve required data from common password excel sheet.

5.1. Retrieve the Password column,



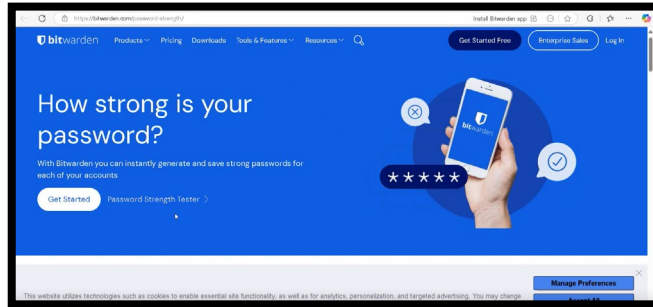
Section 4: check the strength of each password

6.

Access to bit warden portal.

6.1. Access to bit warden portal by URL:

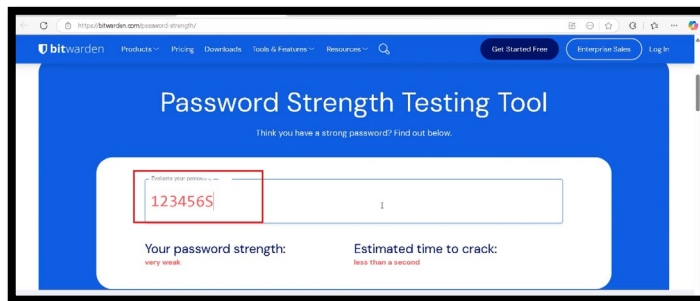
["https://bitwarden.com/password-strength/"](https://bitwarden.com/password-strength/)



7.

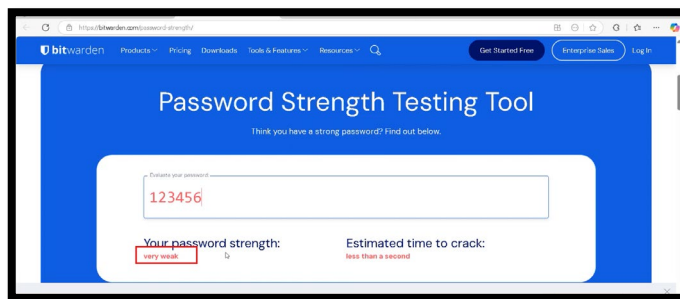
Retrieve required data from portal.

7.1. Scroll down, in evaluating your password fields, enter the password from excel sheet.

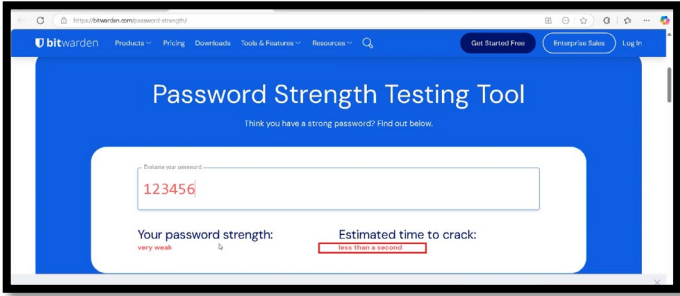


7.2. Click enter button.

7.3. Retrieve the strength.



7.4. Retrieve the estimation time.

		
Section 5: update the report		
8.	Update the report. 8.1. In strength column enter the strength Value. 8.2. In estimation time column, enter the estimation time value. 8.3. Filter column strength, and retrieve the count of each type 8.4. Add a new sheet rename it with count 8.5. Add two new columns with: 8.5.1. Count 8.5.2. Strength type 8.6. In strength type column, enter each type. 8.7. In the count column, enter the total count of each strength type. 8.8. Save the excel.	
Section 6: share the output with concern dept		
9.	Send an email with below details 9.1. To: ###@gmail.com 9.2. Subject: [Process Name] - [today date] 9.3. Attachments: Automated Password Strength Analysis and Reporting report 9.4. Body: Dear, Please find the Automated Password Strength Analysis and Reporting report attached. Best Regards,	

III.1 In Scope for RPA

The activities **in scope of RPA**, are listed here:

1. *The steps mentioned in the PDD are in scope for the automation but not necessarily can be automated 100%. (As there would be a difference in the AS-IS and To-Be process).*
2. *Any exceptions to be handled because of automating the above steps would also be in the scope of RPA automation.*

III.2 Out of Scope for RPA

The activities **OUT of scope of RPA**, are listed here:

1. *Any steps or activities not listed in the AS-IS steps of the PDD are out of scope for the automation.*
2. *Any new changes / enhancements required after the PDD signoff would be considered as a separate engagement and would not be a part of the current project scope.*
3. *Any implementations to be done in the AS-IS process to standardize or digitalize the inputs and data would not be a part of the RPA project scope and must be handled by IT team.*
(Ex: Implementation of Microsoft forms, databases etc.)

III.3 Business Exceptions Handling

The Business Process Owner and Business Analysts are expected to document below all the business exceptions identified in the automation process. These can be classified as:

the description and action to be taken, for each, by the Robot.

Errors identified in the automation process can be classified as:

Known Error or Exceptions

The table below reflects all the errors identifiable in the process evaluation and documentation.

For each of these errors or exceptions, define a corresponding expected action that the robot should complete if it is encountered.

BE #	Exception name	Action to be taken
1	If any missing data in the input email	Send a Business Exception email to SME with exception details.

Unknown Errors and Exceptions

For all the other **unanticipated or unknown business (process) exceptions**, the robot should:

Send email notification and error message screenshot attached.

III.4 Application Error and Exception Handling

A comprehensive list of all errors, warnings or notifications should be consolidated here with the description and action to be taken, for each, by the Robot.

AE #	Exception name	Action to be taken
1	all application failed to launch	Bot retries to Launch the Target Application 3 times until the Applications are opened properly, otherwise, the Bot stops execution and notify the SME with the screenshot.

III.5 Testing Strategy

This section outlines the overall plan and methodology for testing the RPA process. It describes how the testing will be conducted, what types of testing will be used (e.g., unit testing, integration testing, user acceptance testing), and the tools or frameworks involved.

Objective (Goal)	Type of Testing	Responsibilities
- Validation of Functionality. - Identification of Defects. - Verification of Requirements. - Assessment of Process Stability - Usability and User Experience - Risk Mitigation. - Preparation for Deployment. - Security and Compliance Verification	Internal Testing	BA
		Developer
		Team Leader
		Support team
	<u>UAT</u>	SME
		BA
		Developer
		Team Leader
		Support team

III.6 Test Data Requirement Table

The table below outlines the specific test scenarios identified for the process. Each scenario is detailed with the required input data, the expected output upon execution, and an indication of whether specific test data is necessary for the successful completion of the process. This table serves as a guideline for test data preparation and validation to ensure the RPA solution performs as expected in a controlled environment before deployment.

TS #	Test Data Required?	Test Scenario	Input Data (Test Data)	Expected Output	Comments/Note
1	Yes	Happy scenario	Receive input email with required details	Successful	
2	No	Unhappy scenario	Doesn't receive input email with required details	Failed	

III. Other Observations

END OF THE DOCUMENT